Mechanical & Volumetric Design

Material: **GLASS_LINED**
Total Internal Volume: **4.418 m³**
Working Volume at Partial Level: **2.651 m³**
Calculated Shell Thickness: **6 mm**
Empty Reactor Weight: **725.94 kg**

Kinetic Assessment

Required Reaction Volume: **0.076 m³**
Volume Capacity Check: **Yes**

Thermal & Heating/Cooling Setup

Primary Mode: **Heating**
Sensible Fluid Duty: **92.33 kW**
Environmental Heat Transfer (Loss/Gain): **1.04 kW**

Total Compensated Duty: 93.37 kW

Required Utility Mass Flow (Liquid): **15419.6 kg/hr**
Required Utility Pipe ID: **60.3 mm (2.37 inches)**
Required Insulation Thickness: **2.9 mm**

Agitator & Internals Design

Calculated Impeller Diameter: **1.35 m**
Required Motor Power: **14.77 kW (19.81 HP)**
Minimum Shaft Diameter: **54.27 mm**
Estimated Shaft Length: **1.65 m**

Anti-Swirl Baffles

Quantity: **4 (Spaced 90° apart)**
Baffle Width: **150 mm**

Vibration Analysis

1st Critical Speed (Nc): **165.5 RPM**
Safety Check: **Safe (Rigid Shaft Design)**

Reflux Condenser Design

Condenser Heat Duty: **11.67 kW**
Required Cooling Water Flow: **1004.78 kg/hr**
Required Heat Transfer Area: **0.34 m²**

Tube Layout & Shell

Total Number of Tubes: **3**
Estimated Bundle Diameter: **43.31 mm**
Shell Inner Diameter: **63.31 mm**
Required Shell Thickness: **6 mm**

Nozzles

Vapor Inlet ID: **18.81 mm**
Liquid Outlet ID: **4.7 mm**

Required Reactor Nozzles

- Feed Inlet
- Agitator Nozzle
- Bottom Drain
- Jacket Utility In
- Jacket Utility Out
- Manhole

Economic Estimation (CAPEX)

Reactor Body: **$8,711.32**
Agitator System: **$8,862.56**
Condenser: **$136.28**
Bare Equipment: **$17,710.16**

Total Installed CAPEX: $31,878.28

